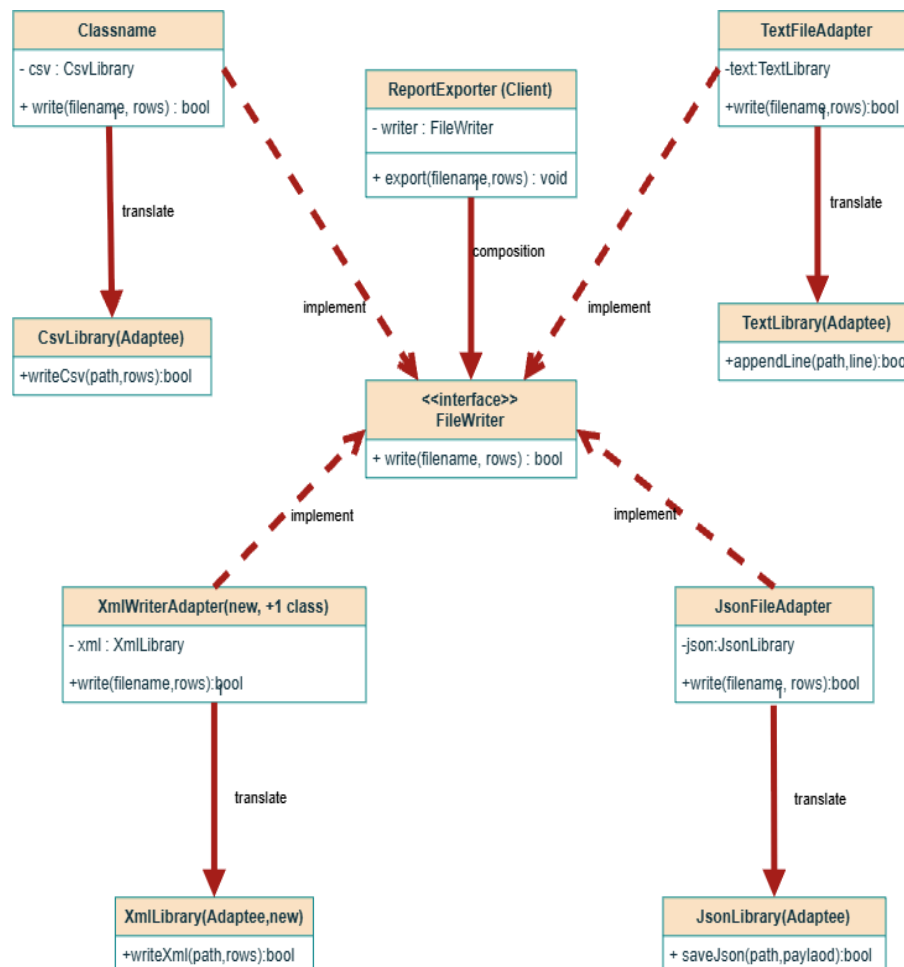




## Lab I: Adapter Reflections

For this lab, I learned how Adapter pattern can help connect classes that have different methods. In the bad version, the code use many if and else statements depending on the file format. It also make the main class handle the writing for CSV, JSON, and text. I think this can make the code more harder to maintain when we add more file types.

For the fixed version, I used a FileWriter interface and made different adapter classes for each file type. The adapters is the one handling the different methods of the old libraries. Because of this, the ReportExporter only need to call the write() method and dont need to know how each file is created.



I also added XML to test if the pattern can be extended. I only needed to create a new XML adapter without changing the ReportExporter. This made me understand why Adapter is useful when we need to use an existing class but the methods does not match what our program needs.

According to Gamma et al. (1994), the Adapter pattern converts the interface of a class into another interface that the client expects. This is what I did in the activity because the different file libraries have different methods, but the adapters make them work using one common interface.